Automated LPG Safety System with “Real-Time Alerts and Absorption Control”

Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

AARMITHA A - 22MIA1138
SURYA M S - 22MIA1098

Dr. PADMANABAN R
School of Computer Science and Engineering

MOTIVATION / INTRODUCTION

- LPG is widely used in homes and industries but poses serious risks if leaked. Traditional leak detection systems are often manual and slow, increasing the danger of fire, explosion, or toxic exposure.
- The system uses MQ-6 gas sensors to continuously monitor LPG levels for quick and accurate leak detection, with real-time data analysis managed by Arduino Uno.
- When a leak is detected, a compartment with zeolites and activated charcoal is automatically activated via a motor (servo, solenoid, or DC) to absorb the leaked gas.
- Communication modules (WiFi, GSM, Bluetooth) provide instant alerts, while weight sensor (HX711) monitor the absorbent’s weight status for timely maintenance.
- The system includes a user-friendly interface with manual override options like a push button or switch for reset and operational control.

OBJECTIVES

- Detect LPG leaks instantly using MQ-6 sensors, and activate an automated absorption unit with zeolites and activated carbon.
- Provide real-time alerts via WiFi, GSM, and Bluetooth, ensuring quick user response even when off-site.
- Monitor absorbent weight with HX711, and offer a user-friendly interface with manual override controls

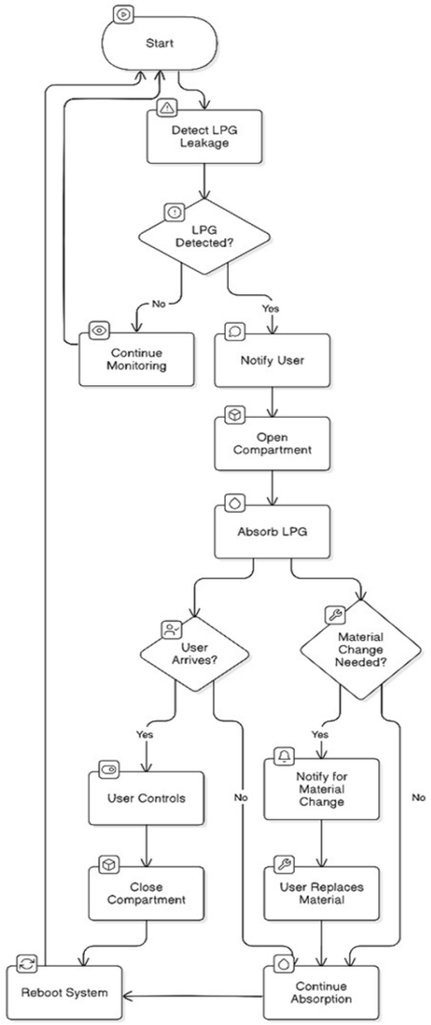
SCOPE OF THE PROJECT

- The project involves designing, building, and testing an automated LPG safety prototype with integrated hardware and sensors.
- It includes software development for data processing, control logic, and real-time alert communication.
- System testing and full documentation cover performance validation, challenges, and future improvement possibilities.

METHODOLOGY

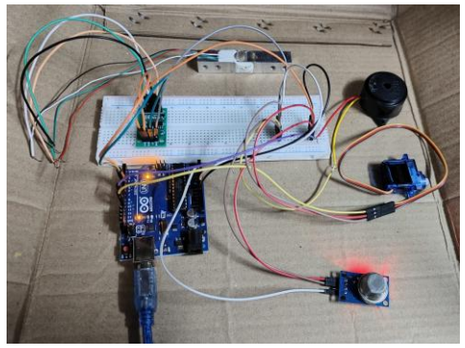
The system continuously monitors for LPG leaks using MQ-6 sensor. On detecting a leak, it sends alerts via Wi-Fi or GSM and activates an absorption unit with charcoal and zeolites. HX711 sensors check absorbent effectiveness, notifying the user when replacement is needed. After user intervention using a push button, the system resets to monitoring mode.

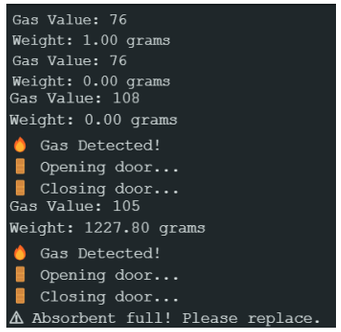
ARCHITECTURE



The above flowchart represents the workflow of how the automated LPG safety system operates.

RESULTS





The prototype successfully detected LPG leakage using the MQ-6 sensor, triggering alerts and automatically opening the absorption compartment via a servo motor. The system displayed gas values above the threshold (e.g., 108, 105), activated the buzzer, and performed door operations as expected. The HX711 sensor effectively monitored the absorbent weight, issuing a "⚠ Absorbent full!" alert when it exceeded the set threshold (e.g., 1227.80 grams). The system reset after each cycle, ensuring continuous monitoring. These results confirm reliable detection, absorption activation, and maintenance alerts, meeting all functional objectives.

CONCLUSION

The project successfully demonstrates an automated LPG leakage detection and safety system. It mainly focuses on quick gas detection, automatic absorption, and real-time user alerts. With added absorbent monitoring and manual control, the system ensures improved safety and timely maintenance.

REFERENCES

1. Sitan, Tamil & Ghafar, Aimi. (2018). Liquefied Petroleum Gas (LPG) Leakage Detection and Monitoring System. Journal of Science and Technology. DOI:10.30880/jst.2018.10.03.007
2. G. N. Sai, K. P. Sai, K. Ajay and P. Nuthakki, "Smart LPG Gas Leakage Detection and Monitoring System," *2023 5th International Conference on Smart Systems and Inventive Technology (ICSSIT)*, Tirunelveli, India, 2023 . **DOI:** 10.1109/ICSSIT55814.2023.10060970